REBECCA M. M. HICKE

PHD CANDIDATE, COMPUTER SCIENCE, CORNELL UNIVERSITY

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EDUCATION

Cornell University, Ph.D. Candidate, Computer Science — Fall 2022 – present
Minor: Information Science — *Ithaca, NY*
GPA: 4.00
Committee: David Mimno (chair), Lillian Lee, Jon Kleinberg, Ross Kristensen-McLachlan

Cornell University, M.S., Computer Science — October 2024
Minor: Information Science — *Ithaca, NY*
Committee: David Mimno (chair), Lillian Lee, Jon Kleinberg

Carleton College, B.A. *cum laude*, Computer Science and English — June 2022
Minor: Digital Arts and Humanities — *Northfield, MN*
Advisors: Eric Alexander and George Shuffelton
Received distinction in both majors

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant, Cornell University — August 2022 – present
Advisor: David Mimno — *Ithaca, NY*
Areas: Digital humanities, natural language processing, machine learning

AI Research Scientist, AIIL Group, Microsoft Research — November 2025 – January 2026
Advisor: Kiran Tomlinson — *Remote*
Areas: Natural language processing, machine learning, user modeling

Research Intern, AIIL Group, Microsoft Research — May 2025 – August 2025
Advisor: Kiran Tomlinson — *Redmond, WA*
Areas: Natural language processing, machine learning, user modeling

Graduate Research Assistant, Aarhus University — May 2024 – August 2024
Advisor: Ross Deans Kristensen-McLachlan — *Aarhus, Denmark*
Areas: Digital humanities, natural language processing, machine learning

PiTech Research Fellow, Cornell University — June 2023 – August 2023
Host Organization: Good Call NYC — *Remote*
Areas: Data visualization, data processing and analysis

Research Scientist, Authors AI — May 2023 – August 2023
Areas: Natural language processing — *Remote*

Undergraduate Research Assistant, Carleton College June 2020 – May 2022
Advisor: Eric Alexander *Northfield, MN*
Areas: Digital humanities, data visualization, human-computer interaction

Undergraduate Research Assistant, UC Berkeley June 2021 – March 2022
REU – Summer Undergraduate Program in Engineering Research at Berkeley *Remote*
Advisors: Sarah Chasins and Aditya Parameswaran
Areas: Human-computer interaction

TEACHING EXPERIENCE

Lead Instructor, Cornell University June 2026 – August 2026
INFO 1100: Introduction to Programming

Graduate Teaching Assistant, Cornell University August 2023 – present
CS/INFO 3300: Data-Driven Web Applications (1 term)
CS 2110: Object-Oriented Programming and Data Structures (1 term)
CS/INFO 4300: Language and Information (1 term)
Received award for TAing contributions

Program Assistant, Carleton College June 2022 – August 2022
History of Computing Off-Campus Studies Program

Undergraduate Teaching Assistant, Carleton College March 2020 – June 2022
CS 111: Introduction to Computer Science (1 term)
CS 201: Data Structures (4 terms)
CS 202: Math of Computer Science (2 terms)

Residential and Program Assistant, UT Austin June 2018 – July 2020
First Bytes Computer Science Camp (2 sessions)
Code Longhorn Computer Science Camp (1 session)
Robotics Camp (2 sessions)
Cybersecurity Camp (1 session)

PUBLICATIONS

Hicke, Rebecca M. M., Sil Hamilton, David Mimno, and Ross Deans Kristensen-McLachlan. 2026. "Attention Flows: Tracing LLM Conceptual Engagement via Story Summaries." *Preprint.*

Hamilton, Sil, **Rebecca M. M. Hicke**, Mia Ferrante, David Mimno, and Matthew Wilkens. 2025. "Too Long, Didn't Model: Decomposing LLM Long-Context Understanding with Novels." *Proceedings of the 9th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature.*

Hicke, Rebecca M. M., Brian Haggard, Mia Ferrante, Rayhan Khanna, and David Mimno. 2025. "Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs." *Proceedings of the 6th Computational Humanities Research Conference.*

Hicke, Rebecca M. M., Yuri Bizzoni, Pascale Feldkamp, and Ross Deans Kristensen-McLachlan. 2025. "Says Who? Effective Zero-Shot Annotation of Focalization." *Proceedings of the 6th Computational Humanities Research Conference.*

Hicke, Rebecca M. M., Sil Hamilton, and David Mimno. 2025. "The Zero Body Problem: Probing LLM Use of Sensory Language." *Proceedings of the 2nd Conference on Language Modeling.*

Hamilton, Sil, **Rebecca M. M. Hicke**, and David Mimno. 2025. "A City of Millions: Mapping Literary Social Networks at Scale." *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities.*

Hicke, Rebecca M. M. and David Mimno. 2025. "*Looking for the Inner Music*: Probing LLMs' Understanding of Literary Style." *Expanding the Toolkit: Large Language Models in Humanities Research – Computational Humanities Research.*

Hicke, Rebecca M. M. and Ross Deans Kristensen-McLachlan. 2024. "SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory." *Proceedings of the 5th Computational Humanities Research Conference.*
Finalist – Best Short Paper Award

Kristensen-McLachlan, Ross Deans, **Rebecca M. M. Hicke**, Márton Kardos, and Mette Thunø. 2024. "Context is Key(NMF): Modelling Topical Information Dynamics in Chinese Diaspora Media." *Proceedings of the 5th Computational Humanities Research Conference.*

Hicke, Rebecca M. M. and David Mimno. 2024. "[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference Annotation with LLMs." *Proceedings of the 8th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature.*

Hicke, Rebecca M. M. and David Mimno. 2023. "T5 meets Tybalt: author attribution in Early Modern English drama using large language models." *Proceedings of the 4th Computational Humanities Research Conference.*

Hicke, Rebecca M. M. and David Mimno. 2024. "'Let every word weight heavy of her worth': Examining How Women Enact Power in Shakespeare's Comedies through Interactive Speech Pattern Visualizations." In *Computational Drama Analysis: Reflecting Methods and Interpretations*. De Gruyter.

Hicke, Rebecca M. M., Maanya Goenka, and Eric Alexander. 2022. "Word Clouds in the Wild." *2022 IEEE 7th workshop on Visualization for the Digital Humanities (Vis4DH)*.

SERVICE

Workshop Organizer: Natural Language Processing for the Digital Humanities [NLP4DH] (2025)

Session Chair: Computational Humanities Research [CHR] (2023)

Reviewer: CHR Journal (2025), Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature [LaTeCH-CLfL] (2024, 2025, 2026), Digital Humanities [DH] (2024, 2025, 2026), CHR (2023), Open Research Europe (2023), Journal of Computational

Literary Studies (2023)

Graduate Mentor + Organizer:

ACSU Research Night (2023, 2024, 2025)

Organized graduate presenters for outreach events getting students interested in research

Student-Applicant Support Program (2022, 2023, 2024)

Provided advice + feedback on application materials for applicants to Cornell CS PhDs

New Visions in Engineering (2022-2023, Coordinator: 2023-2024)

Mentored high school students interested in futures in engineering + CS research

Bowers Undergraduate Research Experience (BURE) (2023)

Mentored and organized social events for undergraduate research assistants

AWARDS & FELLOWSHIPS

Foundational AI PhD Fellowship, Cornell University — January 2026 – May 2026

Hopcroft Fellowship, Cornell University — 2022 – 2023

Laurence McKinley Gould Prize in Natural Science, Carleton College — May 2022

David John Field Prize, Carleton College — May 2022

The David Pollatsek '96 Prize in Computer Science, Carleton College — May 2022

***Distinction* – Computer Science Senior Thesis**, Carleton College — May 2022

***Distinction* – English Senior Thesis**, Carleton College — April 2022

Thorsen Scholarship, Carleton College — 2020 – 2022

Larson International Fellowship, Carleton College — 2021

PRESENTATIONS

Says Who? Effective Zero-Shot Annotation of Focalization, Quid deinde? New Ideas and Approaches to AI & DH — April 2026

How to Read Books with AI: AI and Narrative Analysis, University of Richmond — March 2026
Invited Talk

Says Who? Effective Zero-Shot Annotation of Focalization, CHR 2025 — December 2025

Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs, CHR 2025 — December 2025

The Zero Body Problem: Probing LLM Use of Sensory Language, COLM 2025 — October 2025
Poster Presentation

SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory, CHR 2024 — December 2024

[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference Annotation with LLMs, LaTeCH-CLfL 2024 — March 2024
Poster Presentation

T5 meets Tybalt: author attribution in Early Modern English drama using large language models, CHR 2023 — December 2023

T5 meets Tybalt: author attribution in Early Modern English drama using large language models, TADA 2023 — November 2023
Poster Presentation

Large Language Models and NER: *Better Results with Less Work*, DH 2023 — July 2023
Poster Presentation

Word Clouds in the Wild, Vis4DH – IEEE Vis — October 2022

Examining How Women Enact Power in Shakespeare's Comedies through Interactive Speech Pattern Visualizations, — September 2022
Workshop on Computational Drama Analysis

Practicum 2.0: An Interactive Tool for Introductory CS Topics, Carleton College — May 2022
Computer Science Undergraduate Senior Thesis Talk

"Let every word weight heavy of her worth", Carleton College — May 2022
English Undergraduate Senior Thesis Talk

Technical Talk – "Let every word weigh heavy of her worth", Carleton College — March 2022
English Undergraduate Senior Thesis Technical Talk

Tools for Tracking Police Misconduct Data, Carleton College — October 2021
Summer Research Presentation

Technical Talk – Tools for Tracking Police Misconduct Data, UC Berkeley — August 2021
SUPERB REU Technical Talk

Virtual Poster Session – **Word Clouds in the Wild**, Carleton College — October 2020

Word Clouds in the Wild, Carleton College — August 2020
Summer Research Presentation